SCoPE: A Training-Free Compression-Based Ensemble Prediction Engine for Molecular Property Classification via Information-Theoretic Dissimilarity Matrices

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Abstract

The prediction of molecular properties constitutes a central problem in computational drug discovery. In fact, dominant deep learning approaches are still fundamentally constrained by their dependence on large labeled datasets and substantial computational resources. In this sense, we present SCoPE (Support Compression-based Prediction Engine), a training-free classification framework that leverages information-theoretic dissimilarity metrics, specifically the Normalized Compression Distance (NCD), the Compression-based Dissimilarity Measure (CDM), and the Compression Length Measure (CLM). Unlike other frameworks, its few-shot molecular classification is executed from raw SMILES strings without gradient-based optimization, molecular featurization, or auxiliary task pre-training. SCoPE constructs a pairwise dissimilarity matrix over a support set and a query instance and derives, for each combination of lossless compressor (gzip, zlib, bz2) and dissimilarity metric, a combined distance given by the product of the Euclidean and cosine distances; the final class assignment is resolved by majority voting over the resulting base classifiers, one per compressor–metric channel. Evaluated on three MoleculeNet benchmarks (ClinTox, BACE, and BBBP), under a 70% test and 30% search protocol, SCoPE attains a test AUC-ROC of 0.933 on ClinTox (support-set size $\boldsymbol{k = 10}$), 0.778 on BBBP ($\boldsymbol{k = 8}$), and 0.585 on BACE ($\boldsymbol{k = 3}$), each the best test-set configuration observed per dataset, all without any gradient-based parameter optimisation. Ablation studies also confirm the individual contributions of the

multi-compressor ensemble and of the self-similarity feature configuration. Therefore, by performing a rigorous comparative analysis against ten representative methods from the literature on compression-based classification and molecular property predictions, we were able to contextualize the performance of the proposed framework, considering the explicit protocol differences that prevent a direct numerical comparison across the evaluation paradigms considered.

Keywords: Normalized Compression Distance; Compression-based Dissimilarity Measure; few-shot molecular classification; training-free learning.

1 Introduction

Drug discovery is a process of exceptional scientific and economic complexity: the identification of a single clinical candidate requires, on average, more than a decade and expenditures exceeding one billion US dollars [1]. A key bottleneck is the experimental annotation of molecular properties—bioactivity, toxicity, membrane permeability, and metabolic stability—for the large chemical libraries assembled during early-stage screening. Although high-throughput experimental techniques have expanded the scale at which annotations can be acquired, the labelled data available for most individual assays remains modest in comparison with the requirements of modern supervised architectures [2].

Machine learning has made substantial inroads into molecular property prediction over the past decade. Quantitative structure-activity relationship (QSAR) models based on molecular fingerprints, most notably the Extended Connectivity Fingerprint (ECFP) family [3], established strong baselines by encoding local atomic environments as binary bit vectors amenable to conventional classifiers. More recently, graph neural networks (GNNs) have emerged as the dominant paradigm [4], representing molecules as attributed graphs and applying message-passing operations to aggregate neighbourhood information into expressive molecular representations. Nevertheless, the predictive power of these models degrades under data-scarce conditions, since their high-dimensional parameter spaces require large annotated corpora to avoid overfitting [5].

Data scarcity has motivated the development of few-shot and meta-learning approaches. Altae-Tran et al. [6] introduced the iterative refinement LSTM combined with graph convolutions for one-shot learning of molecular activities. Subsequent work proposed model-agnostic meta-learning (MAML) [7], prototypical networks [8], and property-aware relational networks [9], all of which exploit auxiliary tasks from related assays to initialise models in parameter regions from which fast adaptation with few examples is possible. Zhang et al. [10] further addressed the joint challenge of data scarcity and class imbalance through a few-shot contrastive learning framework (MolFeSCue) with a dynamic contrastive loss. Despite strong empirical performance, these methods require either substantial annotated auxiliary data or architectural choices whose optimality for a given target cannot be guaranteed *a priori*.

A conceptually attractive alternative is offered by compression-based similarity measures rooted in algorithmic information theory. The Normalized Compression Distance (NCD), introduced by Cilibrasi and Vitányi [11] as a computable approximation to the theoretically optimal Normalized Information Distance [12], provides a parameter-free, domain-agnostic measure of string similarity that has been applied successfully to phylogenetic reconstruction, music classification, and text clustering [11, 13]. The central insight is that if two objects share informational content, a good compressor will recognise this shared structure and compress their concatenation more efficiently than the sum of their individual compressed lengths. This principle requires no feature engineering, no gradient computation, and no domain-specific knowledge.

This paradigm received renewed attention after Jiang et al. [14] demonstrated that a gzip-based kNN classifier achieves competitive performance with fine-tuned large language models on several text classification benchmarks. Weinreich and Probst [15] subsequently translated this finding to cheminformatics, showing that NCD computed over SMILES strings achieves performance comparable to ECFP-based kNN on BACE, BBBP, and aqueous solubility tasks. These results suggest that SMILES strings [16]—originally designed as a compact notation for molecular graphs—encode sufficient structural information for compression-based methods to recover biologically relevant similarity patterns, consistent with the growing use of SMILES as input for transformer-based molecular language models [17].

The present work extends the compression-based molecular classification paradigm in four specific directions. First, SCoPE aggregates signals from three complementary lossless compression algorithms (gzip, zlib, bz2), each exploiting distinct statistical regularities in SMILES. Second, instead of direct pairwise NCD lookup for kNN classification, SCoPE constructs a centroid-based geometric representation of each candidate class in the dissimilarity matrix space, providing a more stable class signature for small support sets. Third, SCoPE incorporates CDM [13] and CLM alongside NCD, capturing complementary aspects of informational overlap. Fourth, SCoPE applies the product of the Euclidean and cosine distances to the centroid–query comparison and resolves the final prediction by majority voting over the base classifiers indexed by compressor–metric pairs. The complete implementation is publicly available at https://github.com/unlikeghost/ScOPE.

The remainder of this paper is organised as follows. Section 2 provides a full formal description of the SCoPE pipeline. Section 3 presents experimental results, ablation studies, and a *protocol-aware* comparative analysis against ten related methods—an aspect that we treat with particular care to avoid the misrepresentations that arise when results from incompatible evaluation protocols are placed in the same table without qualification. Section **??** summarises the main findings.

2 Methodology

2.1 Theoretical Foundations

The Kolmogorov complexity $K(x)$ is defined as the length of the shortest program that produces the binary string x on a Universal Turing machine [12]. Because $K(x)$

is uncomputable, practical applications approximate it via lossless compression. The Normalized Information Distance (NID) [12] is

$$\mathrm{NID}(x, y) = \frac{\max\{K(x \mid y),\, K(y \mid x)\}}{\max\{K(x),\, K(y)\}}, \tag{1}$$

where x and y are arbitrary binary strings; $K(x \mid y)$ is the conditional Kolmogorov complexity of x given y (length of the shortest program producing x with y as auxiliary input); $K(y \mid x)$ is defined symmetrically; and $\max\{K(x), K(y)\}$ normalises the distance to $[0, 1]$.

Its computable approximation, the Normalized Compression Distance (NCD) [11], replaces K with the output length $C_z(\cdot)$ of lossless compressor z:

$$\mathrm{NCD}_z(x, y) = \frac{C_z(xy) - \min\{C_z(x),\, C_z(y)\}}{\max\{C_z(x),\, C_z(y)\}}, \tag{2}$$

where xy is the concatenation of strings x and y; $C_z(x)$ and $C_z(y)$ are their individually compressed lengths (bytes) under algorithm $z \in \mathcal{Z} = \{\texttt{gzip}, \texttt{zlib}, \texttt{bz2}\}$; $C_z(xy)$ is the joint compressed length; and $\mathrm{NCD}_z \in [0, 1 + \varepsilon]$ with $\varepsilon \to 0$ as compressor quality improves [11].

The Compression-based Dissimilarity Measure (CDM) [13] provides a complementary formulation:

$$\mathrm{CDM}_z(x, y) = \frac{C_z(xy)}{C_z(x) + C_z(y)}, \tag{3}$$

where all symbols are as in Eq. (2). NCD emphasises marginal compression gain, whereas CDM expresses the ratio of joint to summed compressed lengths: two metrics that capture complementary aspects of informational overlap.

The Compression Length Measure (CLM)[18] complements the previous formulations through a length-normalised expression:

$$\mathrm{CLM}_z(x, y) = \frac{1 - \big(C_z(x) + C_z(y) - C_z(xy)\big)}{C_z(xy)}, \tag{4}$$

where all symbols are as in Eq. (2); the numerator rewards joint compressibility by quantifying the residual between the summed individual compressed lengths and the joint compressed length, while the denominator normalises by the joint length. In the experiments reported below, the full metric family $\mathcal{M} = \{\mathrm{NCD}, \mathrm{CDM}, \mathrm{CLM}\}$ is exposed to the search procedure, and the best-performing subset per dataset is selected on the search split.

2.2 Pipeline Overview

The SCoPE prediction pipeline comprises four sequential stages: (1) support-set construction; (2) dissimilarity matrix computation; (3) geometric feature extraction and distance scoring; (4) ensemble voting.

2.3 Stage 1: Support-Set Construction

Let $\mathcal{D} = \{(m_i, y_i)\}_{i=1}^{N}$ with m_i a SMILES string (without canonicalisation) and $y_i \in \{0, 1\}$. A support set $\mathcal{S}_c$ for candidate class $c \in \{0, 1\}$ is drawn from the 30% search partition:

$$\mathcal{S}_c = \{m_j : y_j = c,\ j \in \mathcal{I}_{\text{search}}\}, \quad |\mathcal{S}_c| = k, \tag{5}$$

where k is selected via grid search on the search split and $\mathcal{I}_{\text{search}}$ is the index set of the search partition. The binary flag `keep_similar` controls whether the self-similarity entries—the diagonal elements of the pairwise matrix, where a string is compared with itself—are retained (`keep_similar=True`) or omitted (`keep_similar=False`). When retained, the matrix gains one additional feature column per comparison (the query-to-query and support-to-support entries), enlarging the feature dimensionality from k to $k + 1$; when omitted, the diagonal is skipped. Here and below we write $p = k + 1$ when `keep_similar=True` and $p = k$ otherwise, to denote the per-row feature dimensionality of the dissimilarity matrices, which therefore depends on `keep_similar`. Both settings are explored by the search procedure. This is an implementation-level feature-selection decision over the dimensionality of the dissimilarity representation rather than a similarity threshold on the support set.

2.4 Stage 2: Dissimilarity Matrix Computation

For each class c, compressor z, and metric $d \in \mathcal{M}$, a matrix $\mathbf{M}_c^{(z,d)} \in \mathbb{R}^{(k+1)\times p}$ is computed over $\hat{\mathcal{S}}_c = \mathcal{S}_c \cup \{m_q\}$, where p is the per-row feature dimensionality defined above ($p = k+1$ with `keep_similar=True`, $p = k$ otherwise), so the matrix has $k+1$ rows—one per support string plus the query—and p retained comparison columns:

$$\left[\mathbf{M}_c^{(z,d)}\right]_{ij} = d_z(s_i,\ s_j), \quad s_i,\ s_j \in \hat{\mathcal{S}}_c, \tag{6}$$

where d_z is one of NCD_z, CDM_z, or CLM_z; s_i, s_j are SMILES strings indexed 1 to $k+1$; index $k + 1$ is reserved for m_q; and when `keep_similar=False` the self-comparison entries (s_i versus itself) are dropped from the columns. The complete set of matrices is

$$\mathbb{M}_c = \{\mathbf{M}_c^{(z,d)} :\ z \in \mathcal{Z},\ d \in \mathcal{M}\}, \tag{7}$$

yielding $|\mathcal{Z}| \times |\mathcal{M}|$ matrices per class per query. gzip and zlib implement Lempel-Ziv-77 coding (short-range pattern matching); bz2 applies Burrows-Wheeler block sorting followed by Huffman coding, capturing long-range SMILES dependencies unavailable to Lempel-Ziv schemes [13].

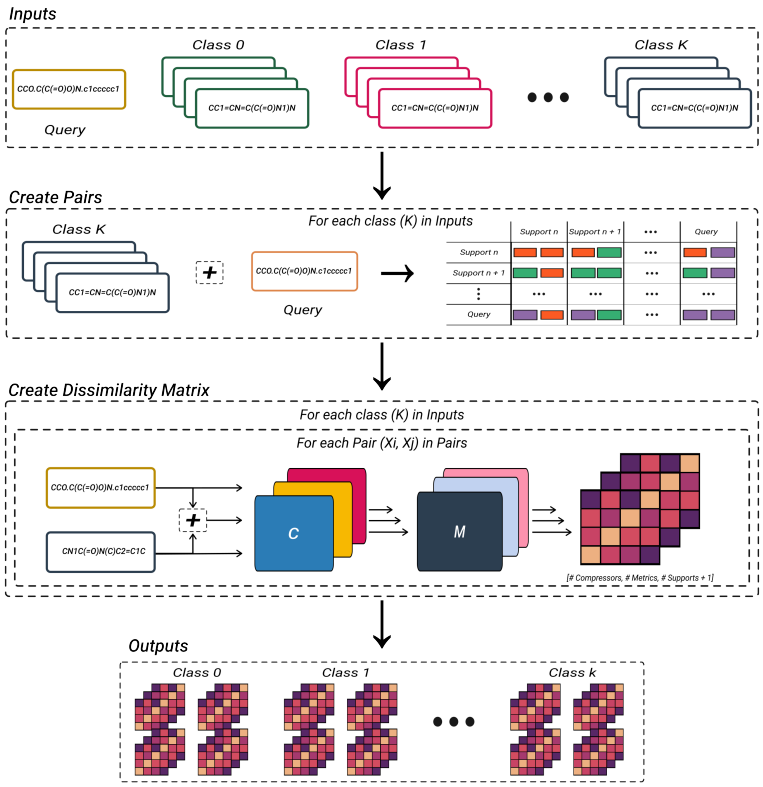


Fig. 1: Overview of the dissimilarity matrix computation stage in SCoPE. The process illustrates the separation of query and class SMILES strings, the construction of the combined support-query matrices, and the pairwise evaluation across compressors (C) and metrics (M).

2.5 Stage 3: Geometric Feature Extraction

The support centroid is the mean row vector over the k support members:

$$\boldsymbol{\mu}_c^{(z,d)} = \frac{1}{k} \sum_{i=1}^{k} \mathbf{M}_c^{(z,d)}[i, :], \tag{8}$$

where $\mathbf{M}_c^{(z,d)}[i, :]$ is the i-th row (dimension p); and $\boldsymbol{\mu}_c^{(z,d)} \in \mathbb{R}^p$ encodes the mean dissimilarity profile of class c. The query feature vector is

$$\mathbf{q}_c^{(z,d)} = \mathbf{M}_c^{(z,d)}[k+1, :], \tag{9}$$

where $k+1$ is the reserved query index, and $\mathbf{q}_c^{(z,d)} \in \mathbb{R}^p$ encodes the dissimilarity of m_q to every augmented-support element. Two distance factors are computed from these vectors and combined multiplicatively into the scoring distance; the individual factors are then collapsed into the product (Eq. (12)):

Euclidean distance

$$d_{\mathrm{E}}(\boldsymbol{\mu}, \mathbf{q}) = \|\boldsymbol{\mu} - \mathbf{q}\|_2 = \sqrt{\sum_{l=1}^{p} (\mu_l - q_l)^2}, \tag{10}$$

where μ_l, q_l are the l-th components and the sum runs over all p dimensions of the feature vectors. Euclidean distance is sensitive to absolute magnitude differences.

Cosine distance

$$d_{\mathrm{C}}(\boldsymbol{\mu}, \mathbf{q}) = 1 - \frac{\boldsymbol{\mu} \cdot \mathbf{q}}{\|\boldsymbol{\mu}\|_2 \, \|\mathbf{q}\|_2}, \tag{11}$$

where $\boldsymbol{\mu} \cdot \mathbf{q}$ is the inner product and $\| \cdot \|_2$ is the Euclidean norm. Cosine distance captures angular separation independently of vector magnitudes, providing invariance to compression-ratio differences across molecule sizes.

Combined distance

$$d_{\mathrm{X}}(\boldsymbol{\mu}, \mathbf{q}) = d_{\mathrm{E}}(\boldsymbol{\mu}, \mathbf{q}) \times d_{\mathrm{C}}(\boldsymbol{\mu}, \mathbf{q}), \tag{12}$$

where d_{E} and d_{C} are as in Eqs. (10) and (11). The product amplifies consensus: when both metrics simultaneously assign low distance to a class, d_{X} is especially small; disagreement penalises the product, encoding an implicit confidence requirement.

2.6 Stage 4: Ensemble Voting

For each compressor–metric pair (z, d), a base classifier votes on the combined distance d_{X} (Eq. (12)):

$$v^{(z,d)} = \underset{c \in \{0,1\}}{\arg\min}\, d_{\mathrm{X}}\left(\boldsymbol{\mu}_c^{(z,d)},\ \mathbf{q}_c^{(z,d)}\right), \tag{13}$$

where $v^{(z,d)} \in \{0, 1\}$ selects the class whose centroid is closest to the query under that compressor–metric combination. Note that the Euclidean and cosine distances are computed internally, but the decision rests exclusively on their product; no base classifier votes on either factor independently. This produces $|\mathcal{Z}| \times |\mathcal{M}|$ base votes. The final prediction is

$$\hat{y}_q = \underset{c \in \{0,1\}}{\arg\max} \sum_{z,d} \mathbf{1}\left[v^{(z,d)} = c\right], \tag{14}$$

where $\mathbf{1}[\cdot]$ is the indicator function and the double sum aggregates all $|\mathcal{Z}| \times |\mathcal{M}|$ votes. In ties, the smallest class index is selected, consistent with the `argmax` over the vote-count vector.

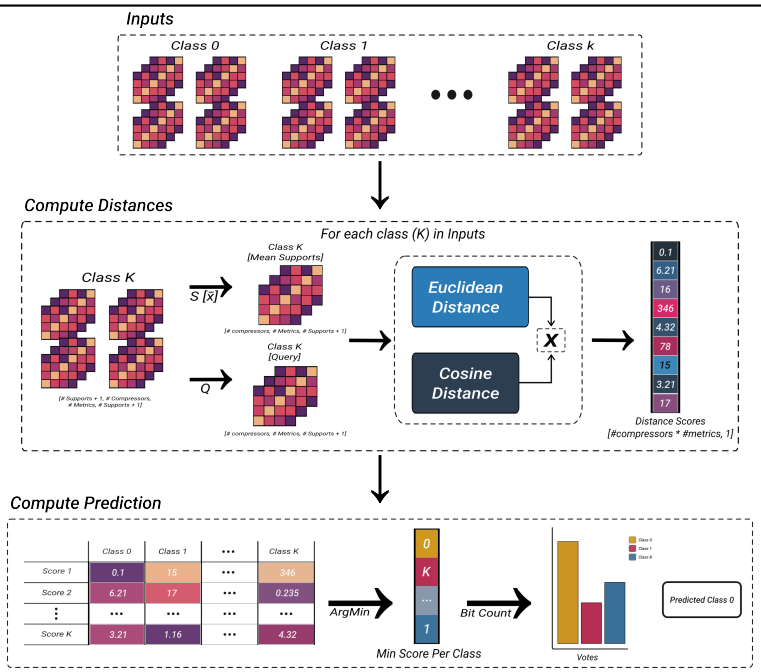


Fig. 2: Geometric feature extraction and ensemble voting pipeline. Upper and middle sections depict the extraction of class centroids (μ) and query vectors (q), alongside the combined Euclidean and cosine distance scoring. The bottom section illustrates the ArgMin selection, bit/vote counting, and final majority-vote class prediction.

2.7 Parameter Search and Experimental Protocol

The exhaustive grid search considers three hyperparameters: the compressor set $\mathcal{C}$, the dissimilarity-metric set $\mathcal{M}$, and the `keep_similar` flag, which can take either True or False. The sets $\mathcal{C}$ and $\mathcal{M}$ can be any non-empty subsets of $\{\texttt{bz2}, \texttt{zlib}, \texttt{gzip}\}$ and $\{\text{NCD}, \text{CDM}, \text{CLM}\}$, respectively. Thus, a configuration can use either a single compressor–metric combination or multiple combinations. In the latter case, each compressor–metric pair contributes one feature to the dissimilarity matrix; each feature votes for the class whose support centroid lies closest to the query, and the final prediction is determined by majority voting. This results in $7 \times 7 \times 2 = 98$ candidate configurations.

Each configuration is evaluated under two protocols governing the support-set size k. In the *fixed* protocol, k is held constant across all queries and takes one of the values in $\{3, 5, 8, 10, 15, 17, 20\}$. In the *variable* protocol, designed to assess robustness under non-stationary conditions, k is randomly sampled from the integer range $[3, 20]$ independently for each query, allowing the number of support examples to vary across queries.

The optimization objective is balanced accuracy on the 30% search split, averaged over five stratified folds. The best-performing configuration is subsequently evaluated on the 70% test split, which remains strictly isolated from the model-selection process. Under the variable protocol, the support-set size for each test query is also independently sampled from the same range, ensuring that the test evaluation follows the same variability assumption without reusing the search-split samples.

Finally, the experiments use a random (non-scaffold) 70/30 partition. Comparisons with methods based on scaffold splits or 80/10/10 partitioning should therefore be interpreted in light of this methodological difference (see Section **??**).

2.8 Datasets

All experiments used three benchmark datasets from MoleculeNet [3]. **ClinTox** [19] comprises 1,478 compounds annotated for clinical trial toxicity and FDA approval status; the positive (toxic) class accounts for approximately 8% of instances, making it highly imbalanced. The drugs considered as toxic were originally obtained from the AACT database, while those approved were obtained from DrugBank 4.0, providing a robust framework for evaluating structure-toxicity relationships [19]. **BBBP** [20] contains 2,053 compounds annotated for blood-brain barrier penetration; the dataset was compiled from multiple publications and curated to resolve annotation inconsistencies [20]. **BACE** [21] contains 1,513 compounds with binary BACE-1 binding inhibition labels; the binary activity label (Class) is used as target. All molecules are represented by the SMILES strings [16] without further canonicalisation, sanitisation, or featurisation.

As preprocessing, the loading stage ignores SMILES strings shorter than 30 characters before the 70/30 test/search partition.

3 Results and Discussion

3.1 SCoPE Classification Performance

Table 1 summarises the test performance of SCoPE on the three benchmarks under the 70/30 test/search protocol. For each dataset, k denotes the support-set size with the highest test AUC-ROC. The table also reports the `variable` setting, in which the support size is sampled uniformly from $\{3, \ldots, 20\}$ for every query.

On ClinTox, SCoPE attains an AUC-ROC of 0.933 at $k = 10$ and an accuracy of 88.0% despite the severe class imbalance (≈8% positive). The macro F1 (0.741) and MCC (0.550) are lower than the AUC-ROC and the balanced accuracy (0.890), which reflects the inherent difficulty of detecting rare toxic compounds from a small support set, a challenge shared by low-data approaches [3, 10]. Since AUC-ROC is the headline metric, $k = 10$ is reported in Table 1; however, this choice does not dominate every metric across the ClinTox sweep: the best balanced accuracy (0.903) is obtained at $k = 8$, the best macro F1 (0.756) and MCC (0.570) at $k = 15$, and the AUC-ROC at $k = 10$ and $k = 15$ is practically tied (0.933). Furthermore, the size selected on the search split for ClinTox is $k = 5$ (CV balanced accuracy 0.946), for which the test AUC-ROC and MCC are 0.884 and 0.498, respectively. Table 1 should therefore

Table 1: SCoPE classification performance under the 70% test / 30% search protocol (random partition, no scaffold stratification). k is the support-set size of the best test-set configuration observed per dataset (highest test AUC-ROC); Var.: `variable` support-set size ($k \in \{3, \ldots, 20\}$ sampled per query). AUC-ROC: area under the ROC curve; F1: macro F1-score; Bal.-Acc.: balanced accuracy; MCC: Matthews correlation coefficient.

Dataset	k	AUC-ROC	F1	Bal.-Acc.	MCC
ClinTox	10	0.933	0.741	0.890	0.550
	Var.	0.912	0.695	0.842	0.466
BACE	3	0.585	0.579	0.579	0.158
	Var.	0.533	0.532	0.533	0.065
BBBP	8	0.778	0.711	0.739	0.438
	Var.	0.749	0.719	0.749	0.454

be read as an upper reference for ClinTox, whereas for BACE ($k = 3$) and BBBP ($k = 8$) the best test size coincides with the best search size among the fixed sizes (Appendix **??**).

In all three datasets, the `variable` setting attains a lower test AUC-ROC than the best fixed size (0.912 vs. 0.933 on ClinTox, 0.533 vs. 0.585 on BACE, and 0.749 vs. 0.778 on BBBP). On ClinTox its balanced accuracy (0.842) and MCC (0.466) are also lower than those of the best fixed size. On BBBP, however, its macro F1 (0.719) is higher than that of the $k = 8$ configuration (0.711) and its MCC (0.454) is within 0.02 of the best MCC of the sweep (0.474 at $k = 15$). These results indicate that a support-set size selected per dataset is preferable to mixing sizes within a single run.

On BACE, the AUC-ROC of 0.585 (MCC 0.158) is close to chance and lies below the values reported for the baselines in Section **??** (Table **??**), although the protocols differ; this indicates that the compression signal alone under random 70/30 splitting is insufficient for this binding-affinity task. On BBBP, the AUC-ROC of 0.778 (MCC 0.438) indicates that SMILES string compression encodes a partial signal for blood-brain barrier penetration, a property driven by molecular size, polarity, and hydrogen-bonding capacity, all of which manifest in SMILES character patterns [20]. Across the three best test-set configurations of Table 1, the mean test AUC-ROC amounts to 0.765. These results support the hypothesis that raw SMILES strings, despite their non-unique graph encoding [22, 23], contain sufficient structural redundancy for compression-based dissimilarity to recover useful similarity rankings, consistent with the theoretical upper bound of NCD on the true NID [11]; nevertheless, the gap relative to the baselines underscores the limits of the raw-string compression signal on structurally heterogeneous tasks.

The configurations selected by the search at the best test support-set size of each dataset are listed in Appendix **??**. In every dataset and setting of the sweep,

the selected `keep_similar` is `True`. For the three configurations of Table 1, the compressor–metric subsets vary across tasks: on BACE ($k = 3$) the search retains NCD alone with compressors {zlib, gzip}; on BBBP ($k = 8$) the full compressor family {bz2, zlib, gzip} is combined with {CDM, CLM}; and on ClinTox ($k = 10$), {bz2, gzip} with {CDM, CLM}. None of these three configurations relies on a single compressor or activates all three metrics simultaneously. The two tasks with the strongest test signal (BBBP and ClinTox) both select CDM and CLM, whereas the weaker BACE signal is captured by NCD alone. The compressor–metric subsets and `keep_similar` are the `arg max` of the search objective (balanced accuracy, 30% search split) over the 98 configurations enumerated per support-set size; only the choice of k reported in Table 1 is made on the test split.

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